## Sine function

1 Consider the function $y = \sin x$.

a Complete the table of values giving answers in exact form:

| $x$ | $0^\circ$ | $30^\circ$ | $90^\circ$ | $150^\circ$ | $180^\circ$ | $210^\circ$ | $270^\circ$ | $330^\circ$ | $360^\circ$ |
|---|---|---|---|---|---|---|---|---|---|
| $\sin x$ | | | | | | | | | |

b Sketch the graph of $y = \sin x$ for $0 \leq x \leq 360$.

c State the sign of $\sin 324^\circ$.

d Which quadrant would the angle $324^\circ$ lie?

e Describe the trend of the graph of $y = \sin x$.

f State the period of $y = \sin x$.

g State the $y$-intercept.

h State the maximum $y$-value.

i State the minimum $y$-value.

j State the range of the function.

2 Consider the function $y = \sin x$.

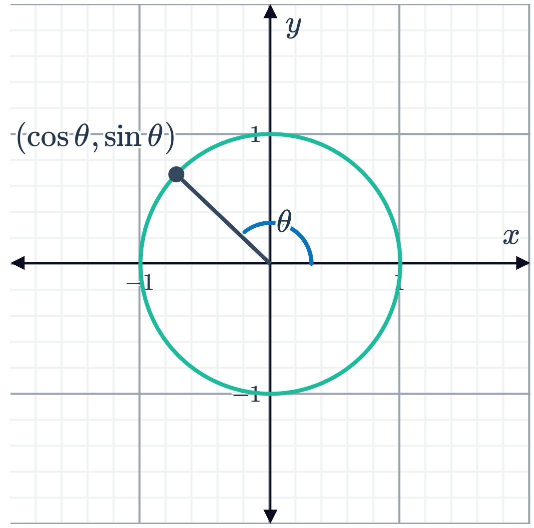
a Complete the table of values giving answers in exact form:

| $x$ | $0$ | $\frac{\pi}{6}$ | $\frac{\pi}{2}$ | $\frac{5\pi}{6}$ | $\pi$ | $\frac{7\pi}{6}$ | $\frac{3\pi}{2}$ | $\frac{11\pi}{6}$ | $2\pi$ |
|---|---|---|---|---|---|---|---|---|---|
| $\sin x$ | | | | | | | | | |

b Sketch the graph of $y = \sin x$ for $0 \leq x \leq 2\pi$.

c State the period of $y = \sin x$.

3 Given the unit circle, explain why the following statements are true for the graph of $y = \sin\theta$:




a The values of $y = \sin\theta$ lie in the range $-1 \leq y \leq 1$.

b The graph of $y = \sin\theta$ repeats after every $360^\circ$.

4 Determine whether the following statements are true of the curve $y = \sin x$:

a The graph of $y = \sin x$ is cyclic.

b As $x$ approaches infinity, the height of the graph approaches infinity.

c The graph of $y = \sin x$ is increasing between $x = -360^\circ$ and $x = -270^\circ$.

d The graph of $y = \sin x$ is symmetric about the line $x = 0$.

e The graph of $y = \sin x$ is symmetric with respect to the origin.

f The $y$-values of the graph repeat after a period of $180^\circ$.

g As $x$ approaches infinity, the graph of $y = \sin x$ stays between $y = -1$ and $y = 1$.

5 Consider the function $y = \sin x$.

a Describe whether the graph is increasing or decreasing in the following intervals:

i $-270^\circ < x < -90^\circ$

ii $-450^\circ < x < -270^\circ$

iii $90^\circ < x < 270^\circ$

iv $-90^\circ < x < 90^\circ$

b Find the $x$-intercept in the following intervals:

i $0^\circ < x < 360^\circ$

ii $-360^\circ < x < 0^\circ$

6 Consider the function $y = \sin x$.

a Describe whether the graph is increasing or decreasing in the following intervals:

i $\dfrac{\pi}{2} < x < \dfrac{3\pi}{2}$

ii $\dfrac{3\pi}{2} < x < \dfrac{5\pi}{2}$

iii $-\dfrac{3\pi}{2} < x < -\dfrac{\pi}{2}$

iv $-\dfrac{5\pi}{2} < x < -\dfrac{3\pi}{2}$

b Find the $x$-intercept in the following intervals:

i $0 < x < 2\pi$

ii $-2\pi < x < 0$

7 The number of people visiting Times Square can be modeled by the function

$$P(t) = 24\,000 \sin \frac{\pi}{12} t + 26\,000$$

where $P(t)$ represents the number of people visiting after $t$ hours.

a State the maximum value of this function over one full cycle.

b State the meaning of the answer from part (a) in context.

## Cosine function

8 Consider the function $y = \cos x$.

a Complete the table of values giving answers in exact form:

| $x$ | $0\degree$ | $60\degree$ | $90\degree$ | $120\degree$ | $180\degree$ | $240\degree$ | $270\degree$ | $300\degree$ | $360\degree$ |
|---|---|---|---|---|---|---|---|---|---|
| $\cos x$ | | | | | | | | | |

b Sketch the graph of $y = \cos x$ for $0 \leq x \leq 360$.

c State the sign of $\cos 340\degree$.

d State the quadrant where an angle with measure $340\degree$ lie.

e State the $y$-intercept.

f State the maximum $y$-value.

g State the minimum $y$-value.

h State the domain of the function.

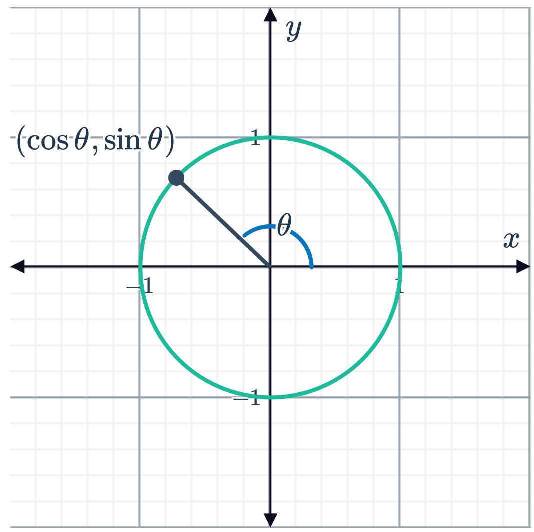
**9** Consider the function $y = \cos x$.

**a** Complete the table of values giving answers in exact form:

| $x$ | $0$ | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ | $\frac{2\pi}{3}$ | $\pi$ | $\frac{4\pi}{3}$ | $\frac{3\pi}{2}$ | $\frac{5\pi}{3}$ | $2\pi$ |
|---|---|---|---|---|---|---|---|---|---|
| $\cos x$ | | | | | | | | | |

**b** Sketch the graph of $y = \cos x$ for $0 \leq x \leq 2\pi$.

**10** Given the unit circle, explain why the following statements are true for the graph of $y = \cos\theta$:




**a** The values of $y = \cos\theta$ lie in the range $-1 \leq y \leq 1$.

**b** The graph of $y = \cos\theta$ repeats after every $360^\circ$.

**11** State whether the following statements are true of the graph of $y = \cos x$:

**a** The graph of $y = \cos x$ is cyclic.

**b** As $x$ approaches infinity, the height of the graph approaches infinity.

**c** The graph of $y = \cos x$ is increasing between $x = 90^\circ$ and $x = 180^\circ$.

**d** The graph of $y = \cos x$ is symmetric about the line $x = 0$.

**e** The graph of $y = \cos x$ is symmetric with respect to the origin.

**f** The $y$-values of the graph repeat after a period of $180^\circ$.

**g** As $x$ approaches infinity, the graph of $y = \cos x$ stays between $y = -1$ and $y = 1$.

12 Consider the function $y = \cos x$.

a If one cycle of the graph of $y = \cos x$ starts at $x = -90^\circ$, where does the next cycle start?

b Describe whether the graph of $y = \cos x$ is increasing or decreasing in the following intervals:

i $-180^\circ < x < 0^\circ$

ii $-360^\circ < x < -180^\circ$

iii $0^\circ < x < 180^\circ$

iv $180^\circ < x < 360^\circ$

c Find the $x$-value of the $x$-intercepts in the following intervals:

i $0^\circ < x < 360^\circ$

ii $-360^\circ < x < 0^\circ$

13 Consider the function $y = \cos x$.

a If one cycle of the graph of $y = \cos x$ starts at $x = -\frac{\pi}{2}$, where does the next cycle start?

b Describe whether the graph of $y = \cos x$ is increasing or decreasing in the following intervals:

i $\pi < x < 2\pi$

ii $-2\pi < x < -\pi$

iii $0 < x < \pi$

iv $-\pi < x < 0$

c Find the $x$-value of the $x$-intercepts in the following intervals:

i $0 < x < 2\pi$

ii $-2\pi < x < 0$

14 The average monthly temperature of a city can be modeled by a cosine graph. Christa has been living in Brooklyn, New York, where the average annual temperature is $55^\circ\text{F}$. She would like to move, and live in a location where the average annual temperature is $72^\circ\text{F}$. When examining the graphs of the average monthly temperatures for various locations, determine whether Christa should focus on each of the following:

a Amplitude

b Midline

c Horizontal shift

d Period

## Tangent function

**15** Consider the equation $y = \tan x$.

**a** Complete the table of values, giving answers in exact form. Write '-' if the value is undefined:

| $x$ | $0^\circ$ | $60^\circ$ | $90^\circ$ | $120^\circ$ | $135^\circ$ | $180^\circ$ | $225^\circ$ | $270^\circ$ | $300^\circ$ | $360^\circ$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $\tan x$ | | | | | | | | | | |

**b** Sketch the graph of $y = \tan x$ for $0 \leq x \leq 360$.

**c** State the $y$-intercept.

**d** State the sign of $\tan 345^\circ$.

**e** State the quadrant where an angle with measure $345^\circ$ lie.

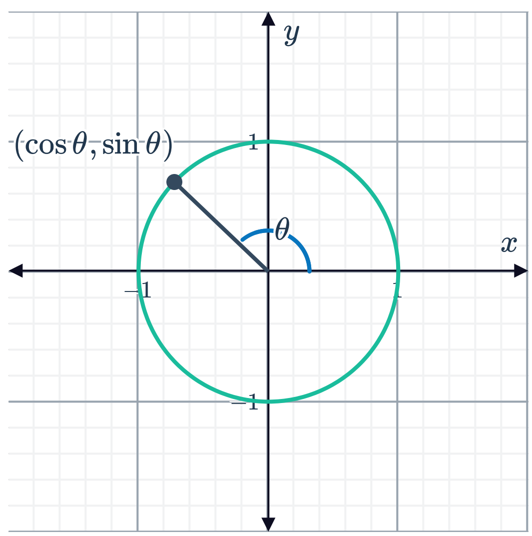
**16** Given the unit circle below, state whether the following statements are true of the graph of $y = \tan x$:

**a** The graph of $y = \tan x$ repeats in regular intervals since the values of $\sin x$ and $\cos x$ repeat in regular intervals.

**b** The graph of $y = \tan x$ is defined for all values of $x$.

**c** Since the radius of the circle is one unit, the value of the function $y = \tan x$ lies in the region $-1 \leq y \leq 1$.

**d** The range of values of function $y = \tan x$ is $-\infty < y < \infty$.




**17** Consider $y = \tan x$ for $-360^\circ \leq x \leq 360^\circ$.

**a** Determine the period of $y = \tan x$ in degrees.

**b** Determine the range of $y = \tan x$.

**c** As $x$ increases, what will be the next asymptote of the graph after $x = 450^\circ$?

18 Consider $y = \tan x$ for $-2\pi \le x \le 2\pi$.

a State the $y$-intercept of the graph.

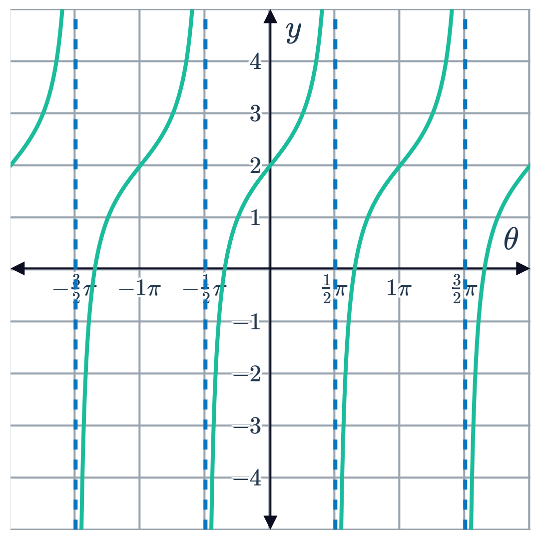
b State the period of the function in radians.

c State the equations of the vertical asymptotes on the domain $0 \le x \le 2\pi$.

d Does the graph of $y = \tan x$ increase or decrease between any two successive vertical asymptotes?

e If $x > 0$, find the least value of $x$ for which $\tan x = 0$.

19 Consider the graph of $y = \tan x + 2$.




a How long is one cycle of the graph?

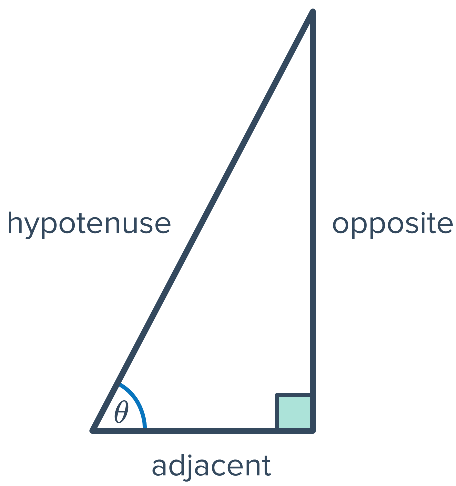
b State the $x$-values for which the equation $\tan x + 2 = 2$, from $x = -2\pi$ to $x = 2\pi$ inclusive.

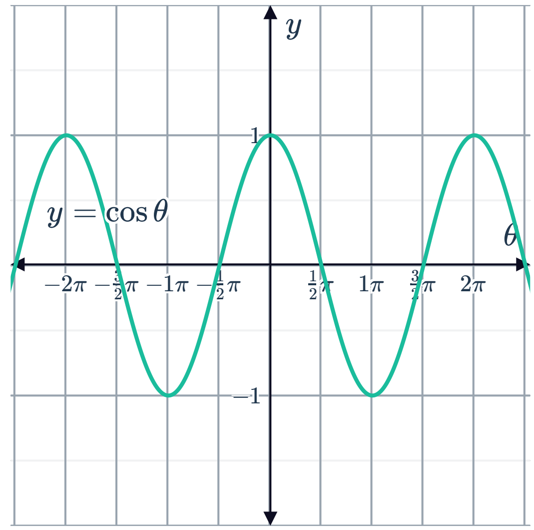
c State the first positive $x$-value for which $\tan x + 2 = 3$.

d Using the period of the graph, for what other values of $x$ between $x = -2\pi$ and $x = 2\pi$ does $\tan x + 2 = 3$?

e For what values of $x$ between $x = -2\pi$ and $x = 2\pi$ does $\tan x + 2 = 1$?

**20** Consider the right triangle containing angle [redacted] the graph of $y = \cos\theta$:




a. As angle $\theta$ increases from $0$ to $\frac{\pi}{2}$, explain what happens to the value of the opposite side in the triangle.

b. Now, explain what happens to the value of $\tan\theta$ as angle $\theta$ increases from $0$ to $\frac{\pi}{2}$, given that $\tan\theta$ is defined as $\frac{\text{opposite}}{\text{adjacent}}$.

c. Express $\tan\theta$ in terms of $\sin\theta$ and $\cos\theta$.

d. Determine the values of $\theta$ for which $\cos\theta = 0$, given $-2\pi \leq \theta \leq 2\pi$.

e. Now, state the values of $\theta$ between $-2\pi$ and $2\pi$ for which $\tan\theta$ is undefined.

f. Complete the tables below:

| $\theta$ | $-2\pi$ | $-\frac{7\pi}{4}$ | $-\frac{5\pi}{4}$ | $-\pi$ | $-\frac{3\pi}{4}$ | $-\frac{\pi}{4}$ |
|---|---|---|---|---|---|---|
| $\tan\theta$ | | | | | | |

| $\theta$ | $0$ | $\frac{\pi}{4}$ | $\frac{3\pi}{4}$ | $\pi$ | $\frac{5\pi}{4}$ | $\frac{7\pi}{4}$ | $2\pi$ |
|---|---|---|---|---|---|---|---|
| $\tan\theta$ | | | | | | | |

g. Now, sketch the graph of $y = \tan\theta$ on the domain $-2\pi \leq \theta \leq 2\pi$.